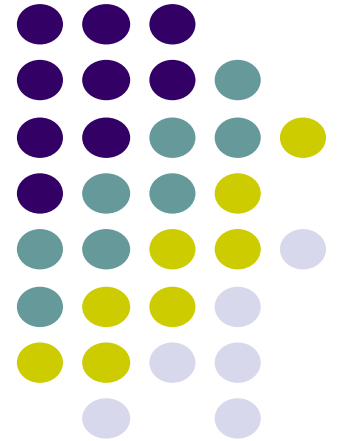
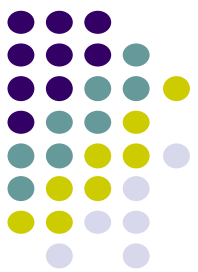


ErSE222: Machine learning in Geoscience

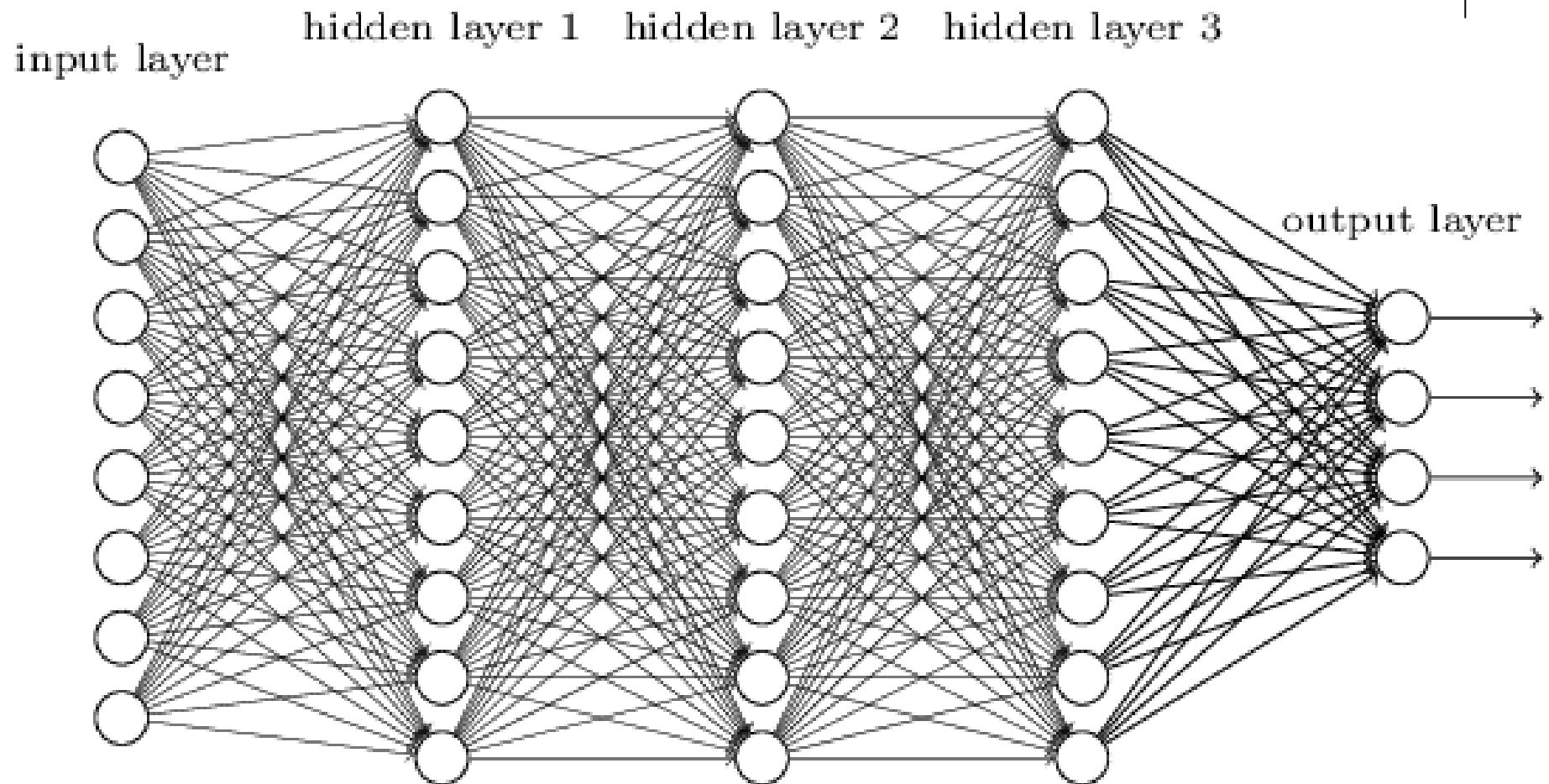
March. 8th, 2026

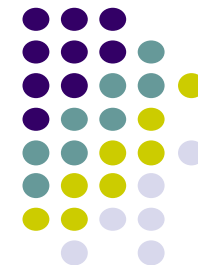
Tariq Alkhalifah and Omar Saad





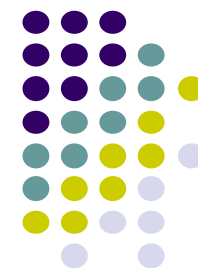
Neural Network





CNN

Convolution



Mathematical operation on two functions (x, h) that keeps track of the running contributions of the product of these two functions

$$y(t) = \int x(\tau)h(t - \tau)d\tau \leftrightarrow y = x * h$$

Diagram illustrating the convolution operation:

- $y(t)$ is labeled as **Output / Feature map** (indicated by an arrow pointing from the text to $y(t)$).
- $x(\tau)$ is labeled as **Input** (indicated by an arrow pointing from the text to $x(\tau)$).
- $h(t - \tau)$ is labeled as **Filter / kernel** (indicated by an arrow pointing from the text to $h(t - \tau)$).

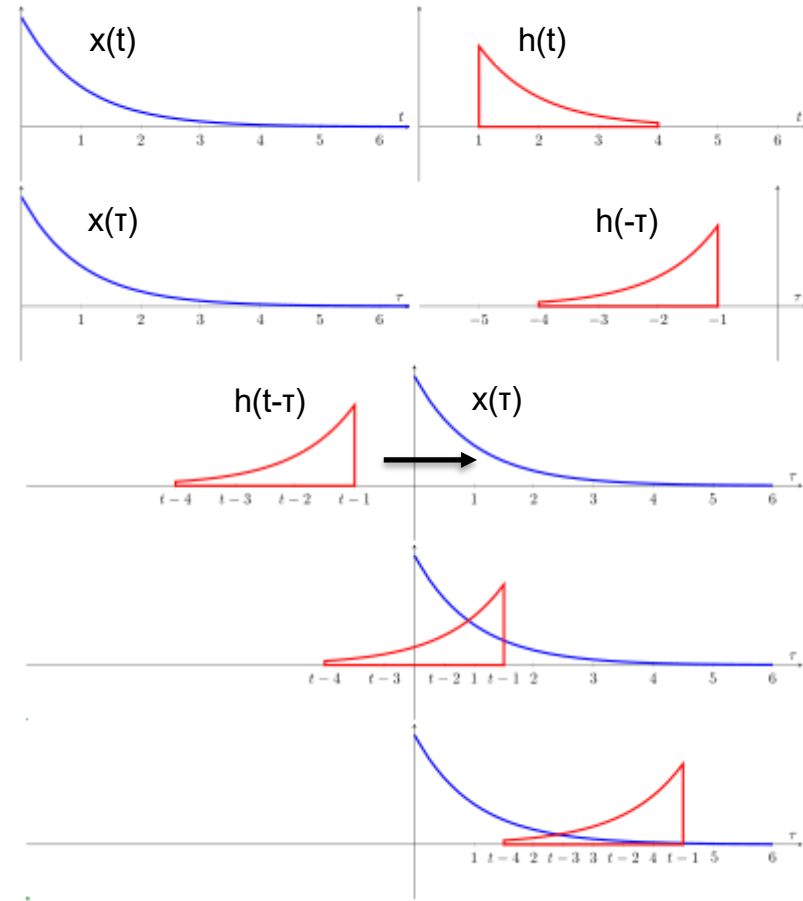
The operation is also represented by the equation $y = x * h$.

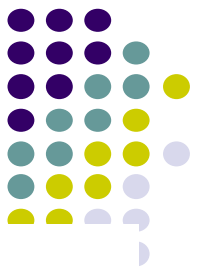
Signal Proc /
DL

Widely used in signal processing, telecommunication, image processing as it describes the response of a system to an input

Convolution (visual)

$$y(t) = \int x(\tau)h(t - \tau)d\tau \leftrightarrow y = x * h$$

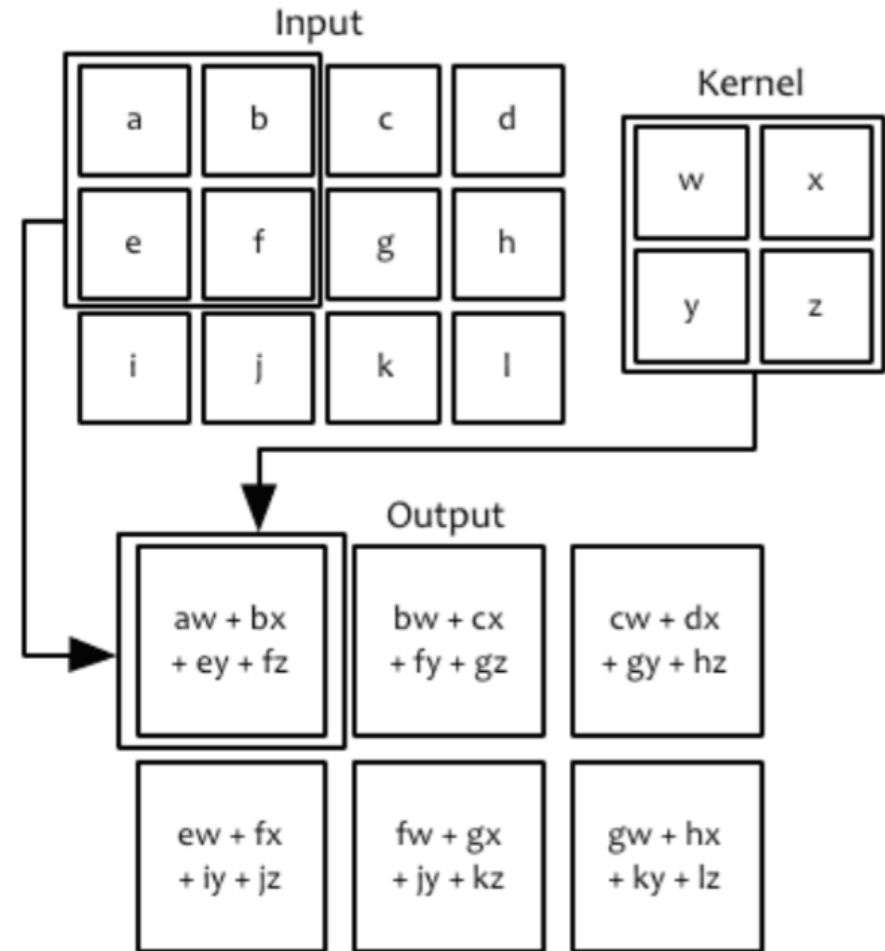




Convolutional Layer

- Input: an image (2-D array): x
- Convolution kernel: W
- Feature map (2-D array of processed data): s
- Convolution operation in 2-D domains:

$$s[i, j] = (x * w)[i, j] = \sum_{m=-M}^M \sum_{n=-N}^N x[i + m, j + n] w[m, n]$$

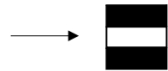
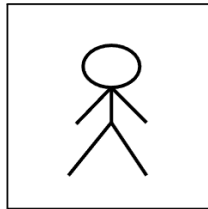


Convolutional Layer

- Sparse interactions or connectivity / weights;



'90



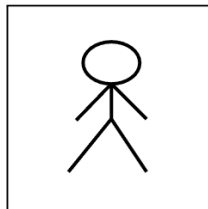
Horizontal edges



Vertical edges

H filters: hand-crafted

'2000

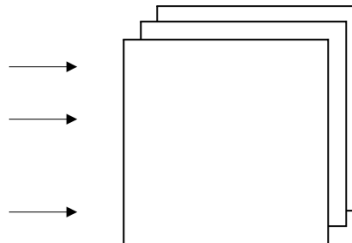


h_1

h_1

...

h_N

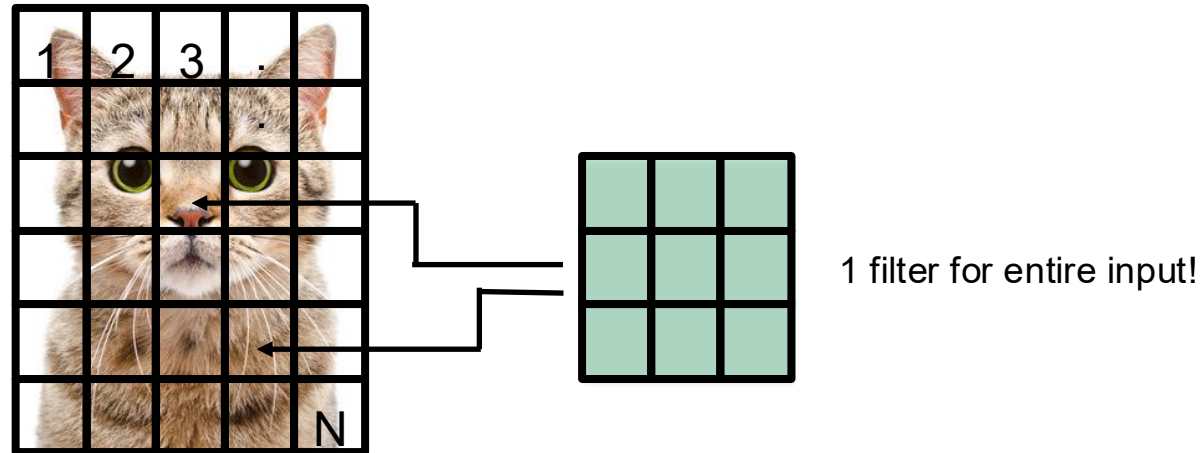


H filters: learned



Convolutional Layer

- Sparse interactions or connectivity / weights;
- Parameter sharing

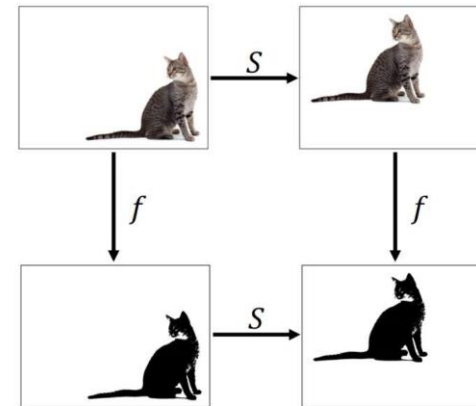




Convolutional Layer

- Sparse interactions or connectivity / weights;
- Parameter sharing
- Equivariance to translation

shift the input by k samples, the output will also be shifted by the same number of samples;





Convolutional Layer (Kernel)

Filter 1

1	-1	-1
-1	1	-1
-1	-1	1

1	0	0	0	0
0	1	0	0	1
0	0	1	1	0
1	0	0	0	1
0	1	0	0	1
0	0	1	0	1

6 x 6 image

3	-1	-3	-1
-3	1	0	-3
-3	-3	0	1
3	-2	-2	-1

1	-1	-1
-1	1	-1
-1	-1	1

1	0	0	0	0
0	1	0	0	1
0	0	1	1	0
1	0	0	0	1
0	1	0	0	1
0	0	1	0	1

6 x 6 image

3	-1	-3	-1
-3	1	0	-3
-3	-3	0	1
3	-2	-2	-1



Convolutional Layer (Padding)

2.0 in ₁	3.0 in ₂	0.0 in ₃	5.0 in ₄	2.5 in ₅	0.0 pad ₁
2.0 in ₆	1.5 in ₇	0.5 in ₈	0.0 in ₉	7.0 in ₁₀	0.0 pad ₂
1.5	5.0	5.0	3.0	2.0	0.0
3.0	5.0	7.0	1.5	0.0	0.0
2.0	5.0	2.0	1.5	2.0	0.0
0.0	0.0	0.0	0.0	0.0	0.0



3.0 out ₁	5.0 out ₂	7.0 out ₃

Convolution and Correlation outputs

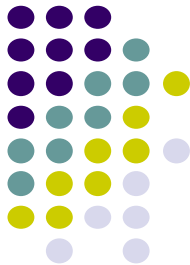


The size of the output of these two operations can be:

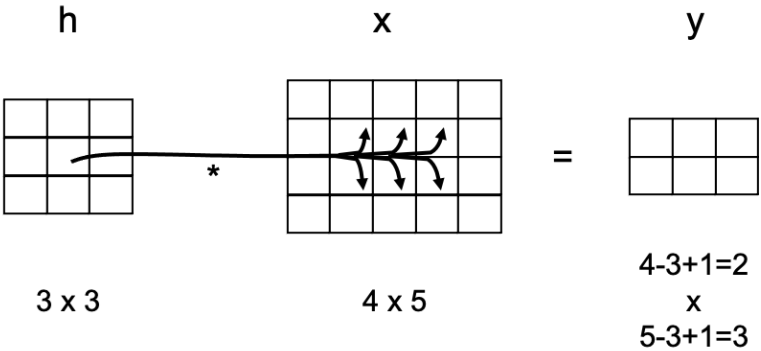
$$N_y = N_x - N_h + 1$$

*when we are interested in the valid signal
(i.e., the entire filter contributes to the output calculation)*

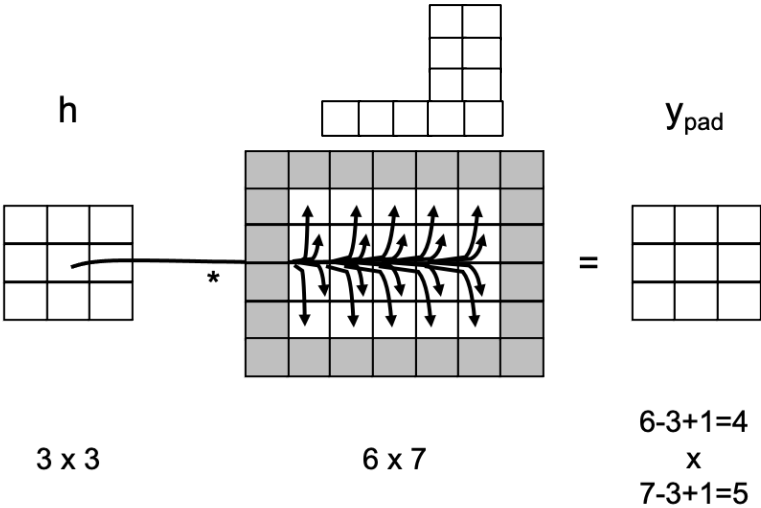
Padding



$$N_y = N_x - N_h + 1$$

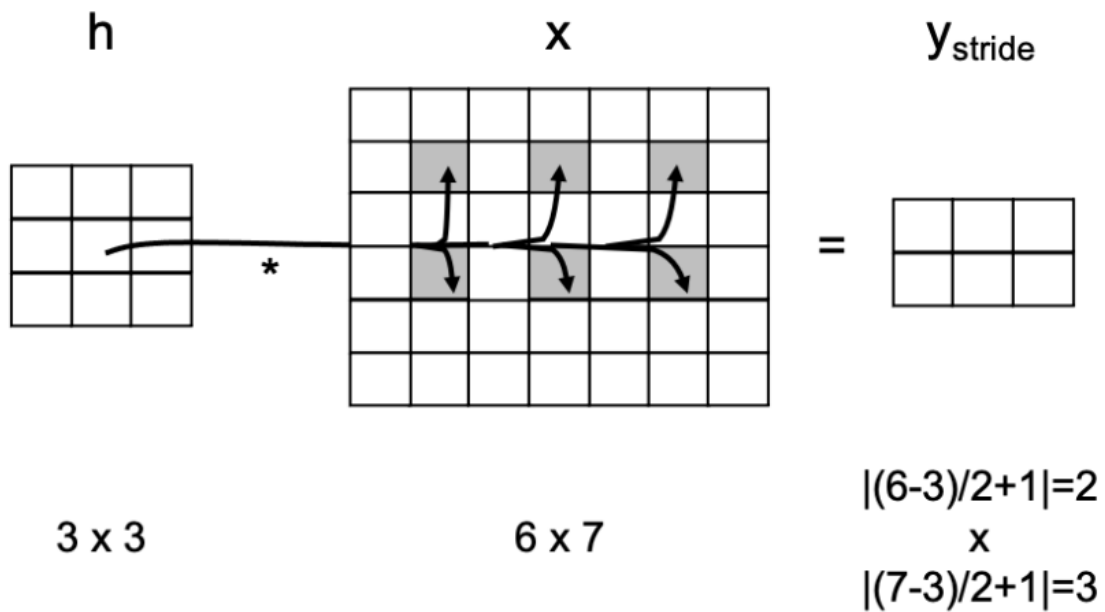


No padding



With padding

Strides



With striding

$$N_y = \left\lfloor \frac{(N_x - N_h)}{stride} + 1 \right\rfloor.$$

Padding and Strides



$$N_y = \left\lfloor \frac{N_x + 2pad - N_h}{stride} + 1 \right\rfloor$$

Convolutional Layer (Pooling)



- Subsampling pixels will not change the object

bird



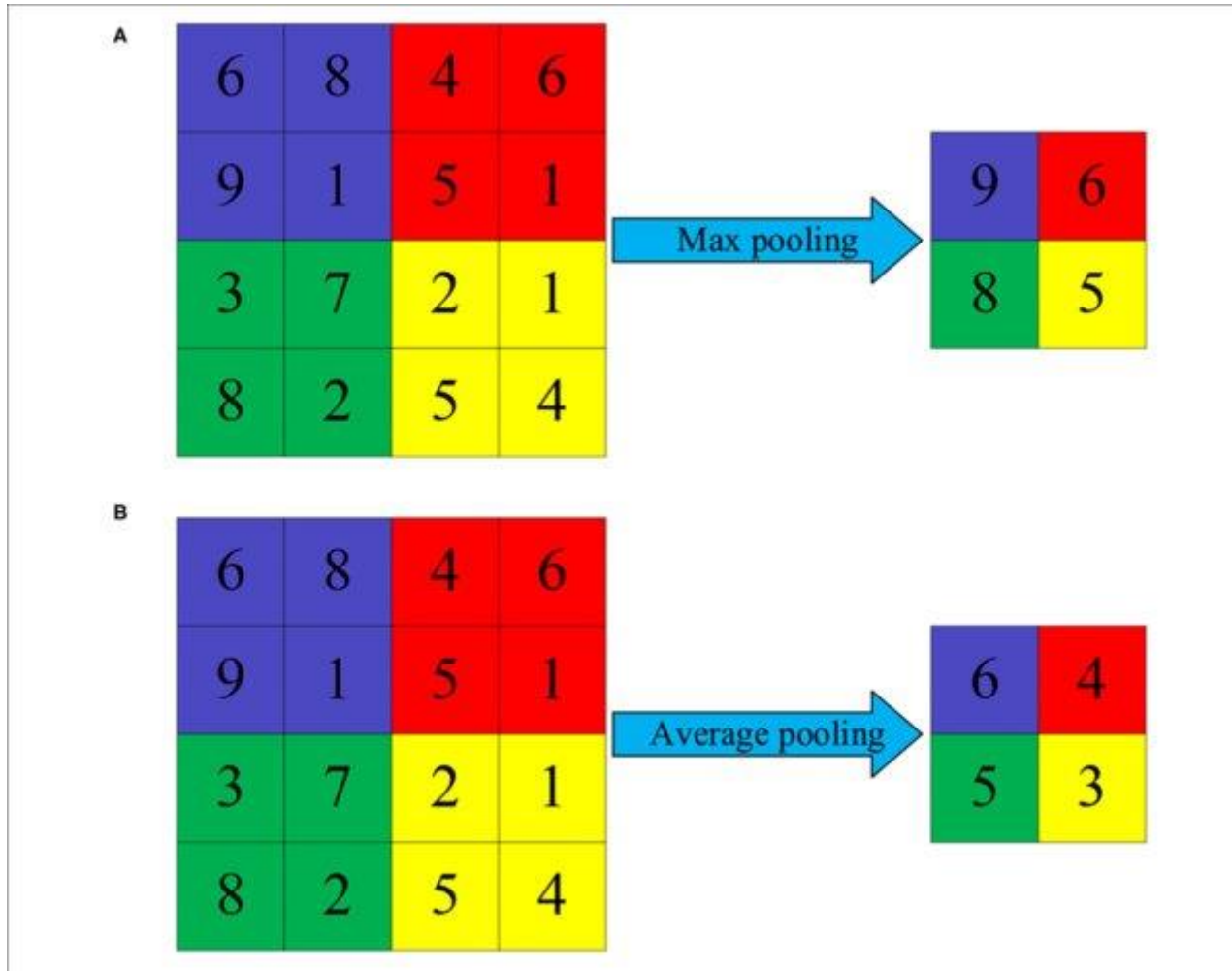
Subsampling

bird

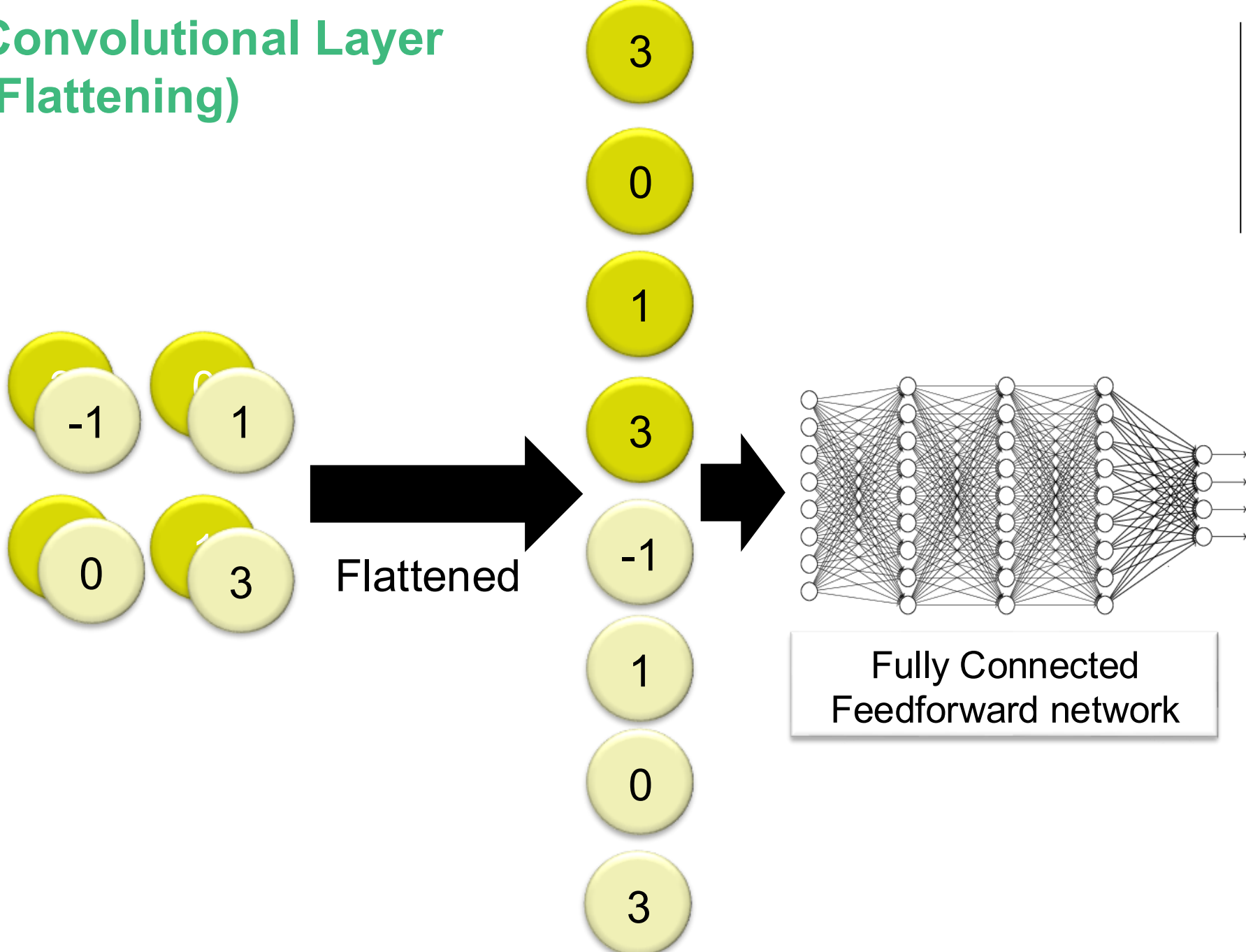


We can subsample the pixels to make image smaller,
fewer parameters to characterize the image

Convolutional Layer (Pooling)



Convolutional Layer (Flattening)





Convolutional Layer (All in one)



Convolution

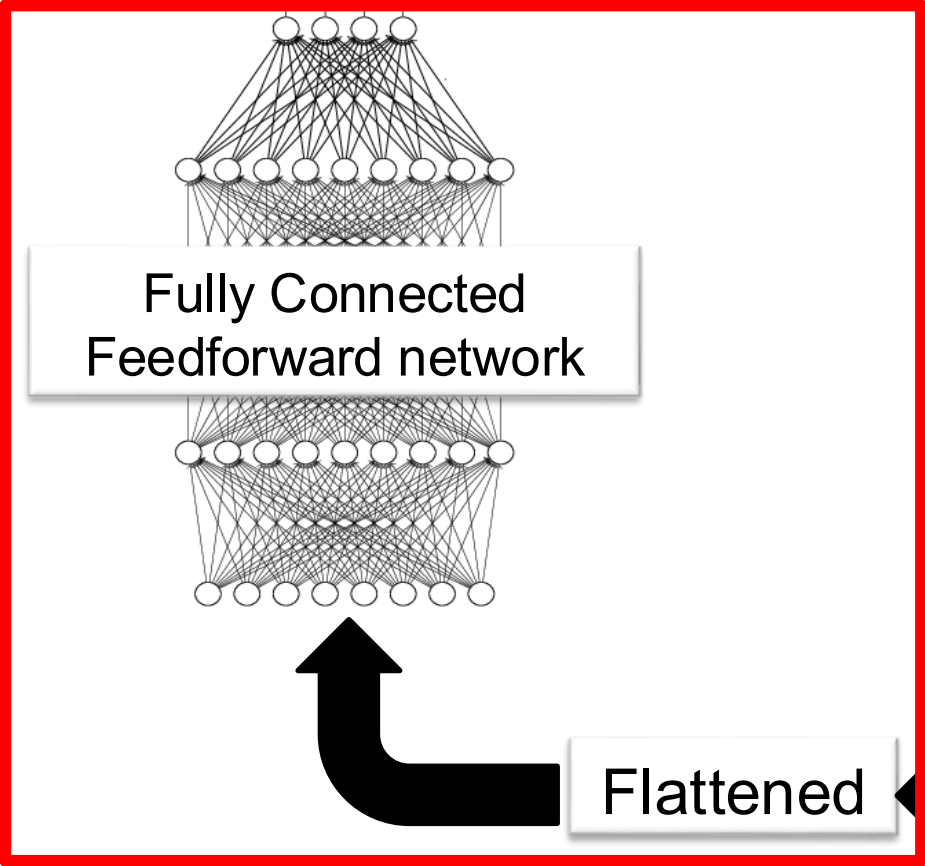
Max Pooling

Convolution

Max Pooling

Flattened

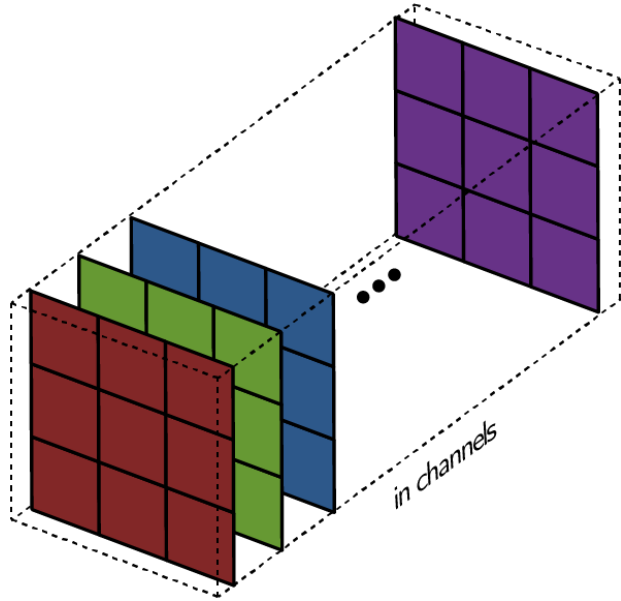
cat dog



Channels



In deep learning, we work with multiple feature maps at the same time



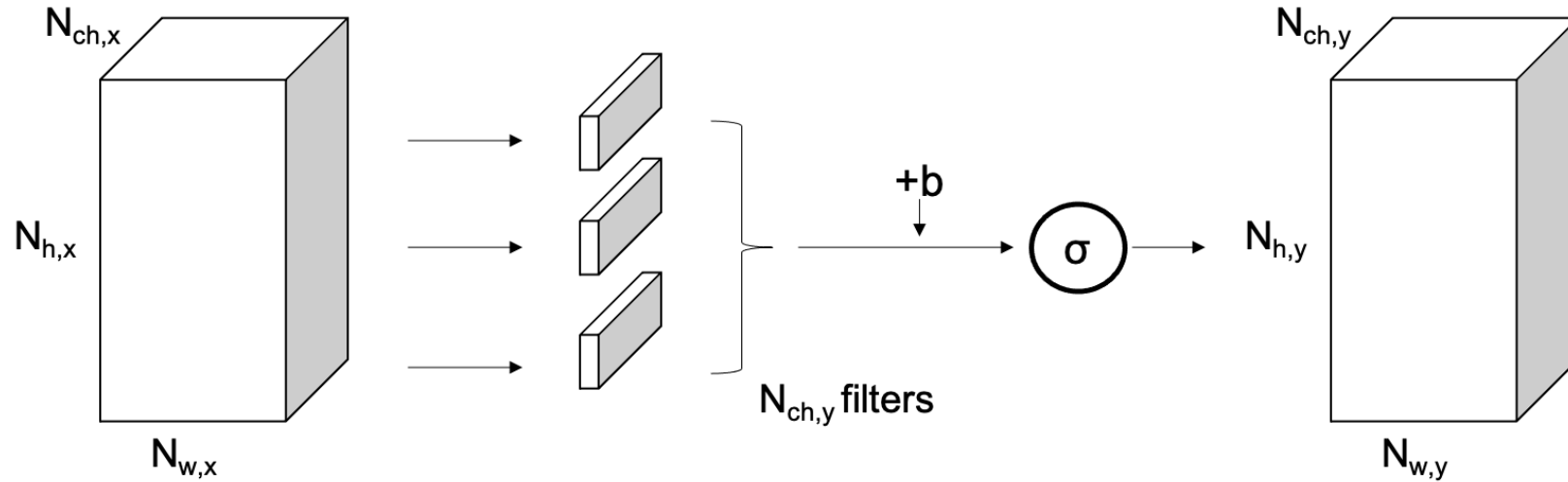
Input: can be RGB, spectral bands, horizontal/vertical motion, etc.

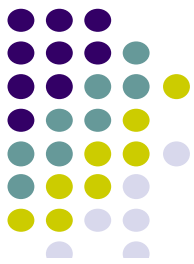
Hidden layers: produced by applying multiple filters in previous conv layer

Convolutional layer

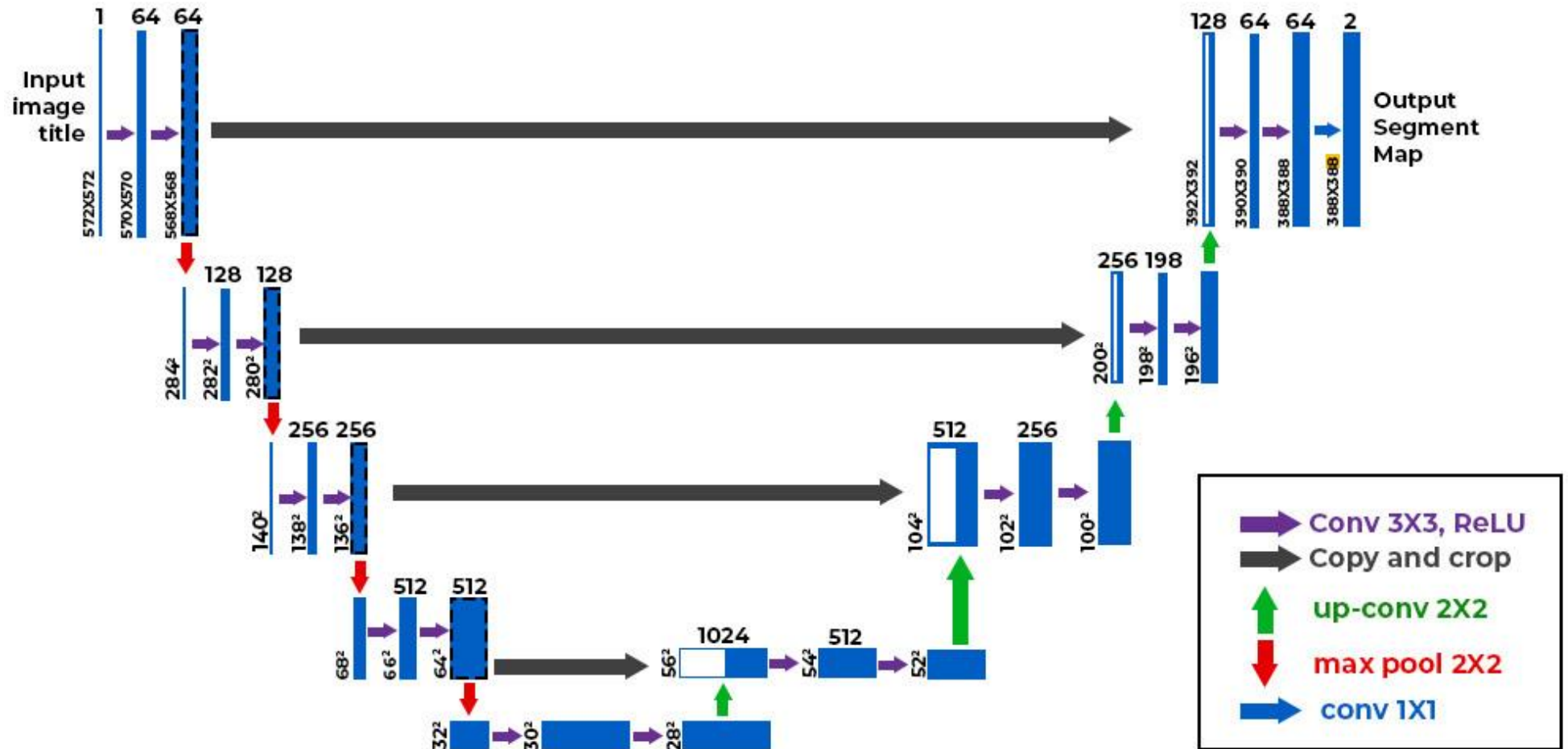


As usual, we can apply multiple filters to produce multiple outputs (like in MLP we used to apply multiple weight vectors = weight matrix)

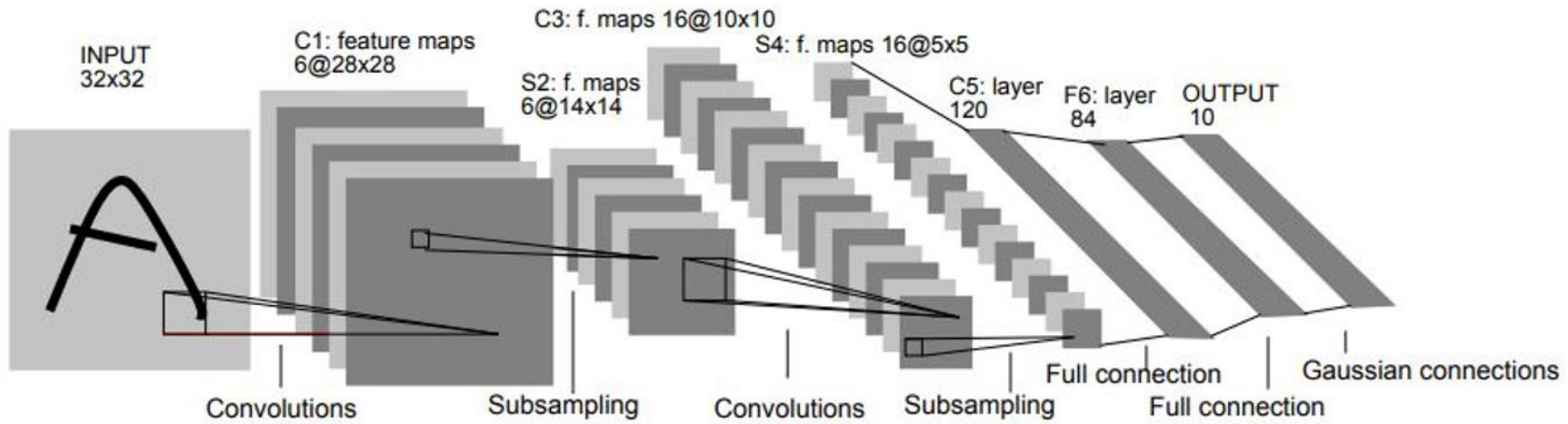




Convolutional Layer (U-NET)

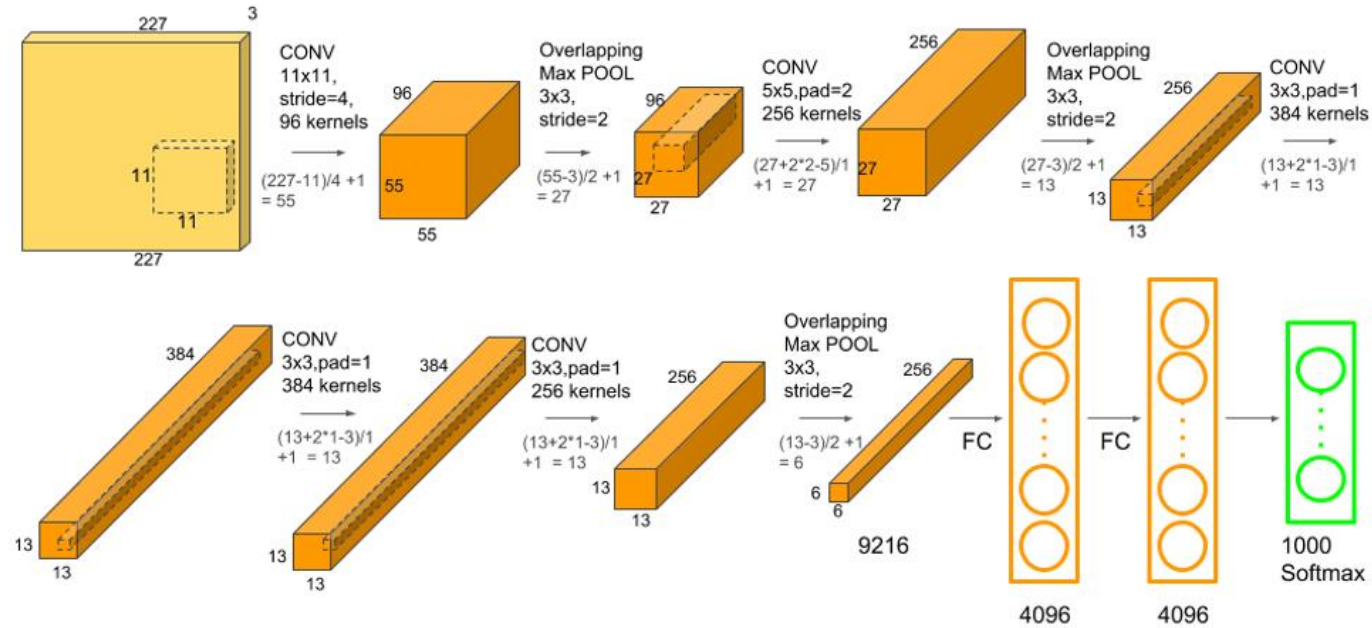


LeNet-5



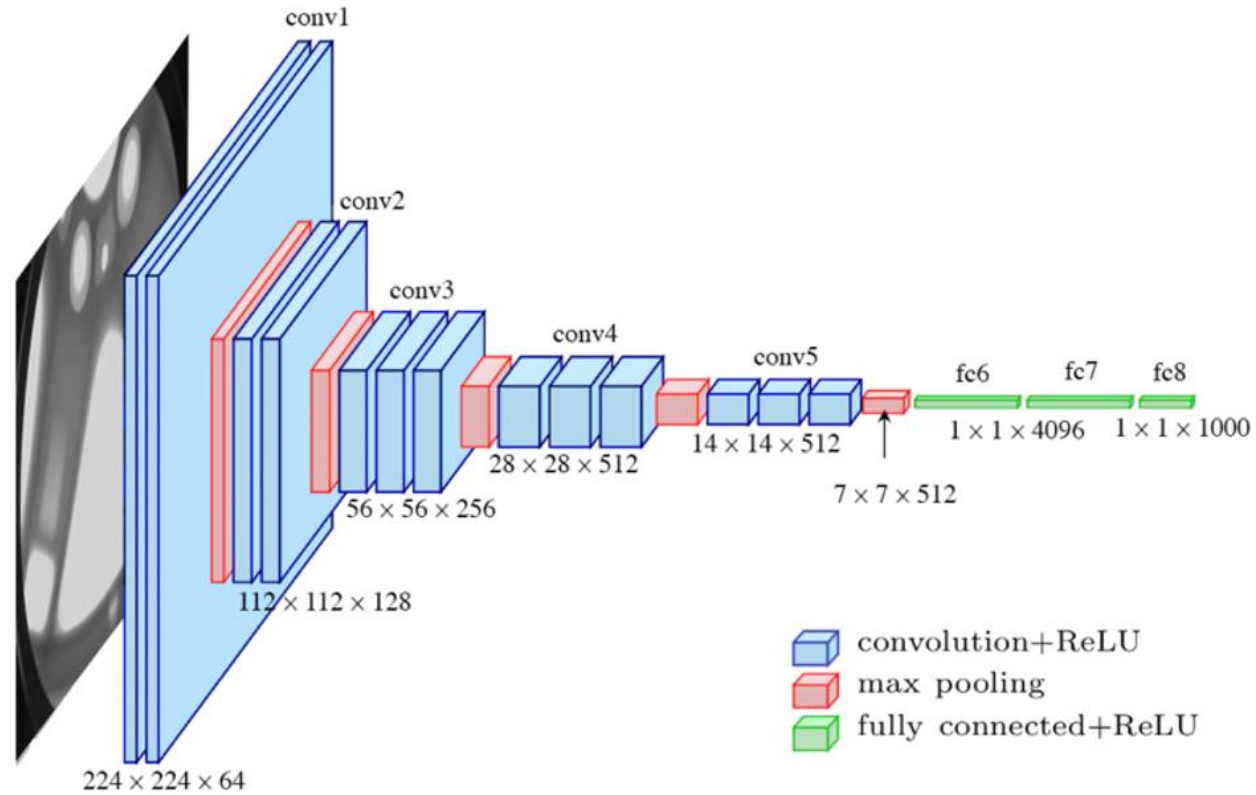
Source: <https://paperswithcode.com/method/lenet>

AlexNet



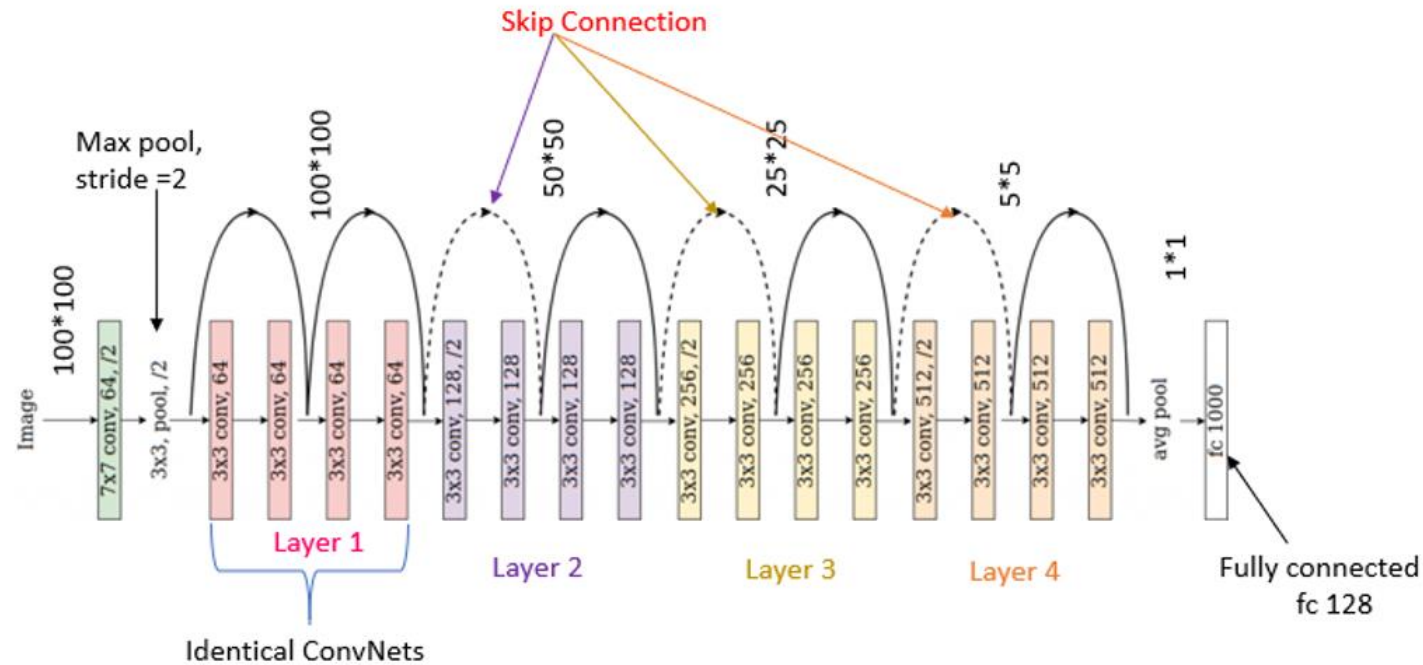
Source: <https://neurohive.io/en/popular-networks/alexnet-imagenet-classification-with-deep-convolutional-neural-networks/>

VGG



Source: https://www.researchgate.net/publication/322512435_Automatic_localization_of_casting_defects_with_convolutional_neural_networks

ResNet



ResNet-18 Architecture

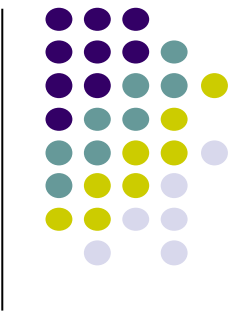
Fruit 360 Input Image size= 100*100 px

Source: <https://www.pluralsight.com/guides/introduction-to-resnet>

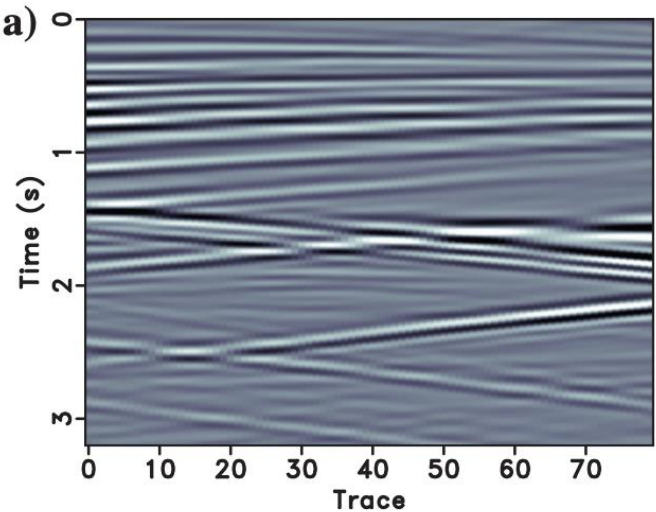


Application

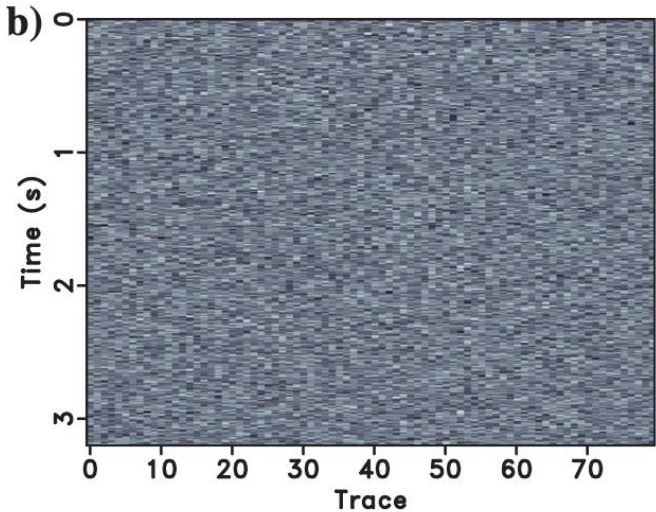
Seismic Data Denoising



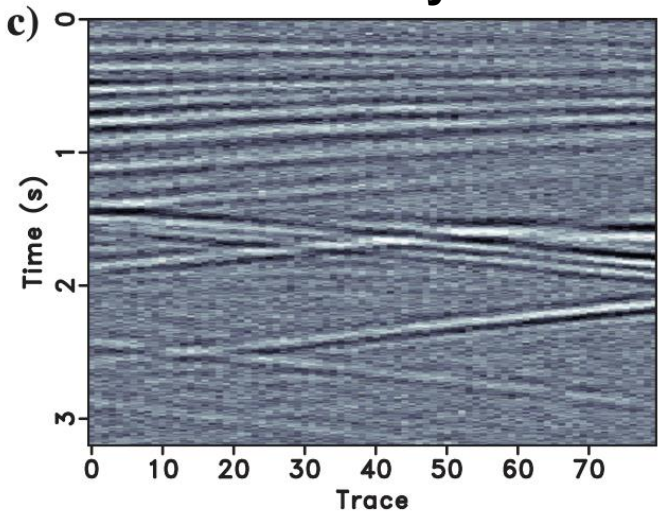
Clean



Noise



Noisy





Thank You

30 17:50